

Yejun Yeo

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EDUCATION

University of Waterloo

B.C.S Computer Science Honours Co-op

Waterloo, Canada

September 2024 – May 2029

- **Awards:** President's Scholarship of Distinction, 2nd Place – 2025 Waterloo Tech Week Chess Bot Tournament
- **Certifications:** Stanford Online Machine Learning, Google Data Analytics Professional Certificate

EXPERIENCE

Machine Learning Engineer

City of Edmonton

Jan. 2026 – Aug. 2026

Alberta, Canada

- Built a machine learning model predicting wildfires in Edmonton using R, chosen for its geospatial libraries, trained on weather and land data pulled from PostgreSQL
- Reduced false alarm ratio from 40:1 to 13:1 (3× precision gain) and improved ROC-AUC from 0.60 to 0.83 by engineering 88+ geospatial features, implementing isotonic probability calibration, tuning XGBoost for extreme class imbalance (0.33% positive rate), and etc.
- Ported the R/Shiny wildfire prediction model to FastAPI + React/TypeScript, including hand-porting R fire prediction libraries to Python, achieving a 6× faster load time (2 min to 20 s)
- Designed a production pipeline with Docker, GitLab CI/CD, and Google Cloud (GKE) under mentorship from senior architects, implementing features based on their reviews
- Collaborated with 3 senior engineers through Git feature branches and merge requests, iterating on the app based on bi-weekly stakeholder feedback

PROJECTS

Brain Tumor MRI Classifier | *Python, PyTorch, OpenCV, NumPy, Matplotlib*

- Built a CNN in PyTorch and Python to classify brain tumour MRI scans from a Kaggle dataset into 4 categories
- Pre-processed images with OpenCV and NumPy by resizing, grayscaling, and normalizing to give the model consistent input
- Tuned the PyTorch architecture with ReLU, max pooling, and softmax layers, and tracked accuracy and loss curves in Matplotlib to catch overfitting

Machine Learning-Based PRNG Research Paper | *Python, Java, NumPy, Git*

- Authored a 4,000-word research paper comparing the predictability of PRNGs (Pseudorandom Number Generators) used in Python and Java
- Implemented linear regression with gradient descent in Python to evaluate error margins across 400×25 datasets of pseudo-random numbers
- Analysed results showing Python's PRNG (Mersenne Twister) is slightly more predictable than Java's LCG (Linear Congruential Generator)

TECHNICAL SKILLS

Programming Languages: Python, R, C/C++, Java, SQL (Postgres), JavaScript, TypeScript, HTML/CSS

Frameworks: React, Node.js, Flask, FastAPI, Streamlit

Developer Tools: Git, Docker, Kubernetes (GKE), Google Cloud Platform, VS Code, Vim

Libraries: NumPy, pandas, Matplotlib, PyTorch

Languages: English (native), Korean(native), German, Slovak